

1. What problem does SQL solve that CSV files cannot?

SQL allows you to efficiently store, organize, search, update, and analyze large amounts of data. Unlike CSV files, SQL databases:

- Handle millions of records efficiently.
- Support multiple users accessing data simultaneously.
- Maintain relationships between tables.
- Enforce data integrity with constraints.
- Allow complex queries, filtering, sorting, and aggregation.

CSV files are simply text files that store data and lack these features.

2. What is the difference between a database table and a spreadsheet?

Database Table	Spreadsheet
Stores structured data in a database	Stores data in a file (e.g., Excel)
Designed for large datasets	Better for small to medium datasets
Supports relationships between tables	Usually standalone data
Multiple users can access simultaneously	Collaboration is more limited
Uses SQL for querying	Uses formulas and manual filtering

3. What is a Primary Key?

A Primary Key is a column (or combination of columns) that uniquely identifies each row in a table.

Example:

StudentID (Primary Key)	Name
101	Ali
102	Sara

No two rows can have the same primary key value, and it cannot be NULL.

4. What is a Foreign Key?

A Foreign Key is a column in one table that refers to the Primary Key in another table. It creates a relationship between tables.

Example:

Students

StudentID	Name
101	Ali
102	Sara

Orders

OrderID	StudentID
1	101
2	102

Here, **StudentID** in the Orders table is the foreign key.

5. What is the difference between WHERE and HAVING?

WHERE	HAVING
Filters rows before grouping	Filters groups after grouping
Cannot use aggregate functions	Can use aggregate functions
Used before GROUP BY	Used after GROUP BY

Example:

```
-- WHERE
```

```
SELECT * FROM Employees
```

```
WHERE Salary > 50000;
```

```
-- HAVING
SELECT Department, AVG(Salary)
FROM Employees
GROUP BY Department
HAVING AVG(Salary) > 50000;
```

6. What is the difference between ORDER BY and GROUP BY?

ORDER BY

Sorts the results

Does not combine rows

Used for presentation

Example:

```
ORDER BY Salary DESC;
```

sorts salaries from highest to lowest.

```
GROUP BY Department;
```

groups employees by department.

GROUP BY

Groups rows with the same values

Combines rows for aggregation

Used with aggregate functions

7. What does DISTINCT do?

DISTINCT removes duplicate values from the result.

Example:

Table:

City

Lahore

Karachi

Lahore

Query:

```
SELECT DISTINCT City  
FROM Customers;
```

Result:

City

Lahore

Karachi

8. When should you use LIMIT?

Use LIMIT when you want to return only a certain number of rows.

Example:

```
SELECT *  
FROM Employees  
LIMIT 10;
```

This returns only the first 10 rows.

Common uses:

- Previewing data.
- Improving query performance.
- Pagination.

9. What are aggregate functions?

Aggregate functions perform calculations on multiple rows and return a single value.

Common aggregate functions:

- COUNT() – Counts rows.
- SUM() – Adds values.
- AVG() – Calculates the average.
- MAX() – Finds the highest value.

- MIN() – Finds the lowest value.

Example:

```
SELECT  
COUNT(*) AS TotalEmployees,  
AVG(Salary) AS AverageSalary  
FROM Employees;
```

10. Why do Data Scientists prefer databases over Excel for large datasets?

Data Scientists prefer databases because they:

- Handle millions of rows efficiently.
- Execute queries much faster.
- Reduce data duplication through normalization.
- Support multiple users simultaneously.
- Provide better security and access control.
- Easily integrate with programming languages like Python and R.
- Can manage data too large to fit comfortably in Excel.

Excel is useful for small datasets and quick analysis, while databases are more suitable for storing, managing, and analyzing large, complex datasets.